

Cores of Binary 3-Qualitative Independence Hypergraphs

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Abstract

Let $3\text{-QI}(n, 2)$ be the 3-uniform hypergraph whose vertices are unordered bipartitions of an n -element set into two blocks of size at least 4. Three such bipartitions form an edge if their common refinement has all eight cells nonempty. We prove that this hypergraph is a core for every $n \geq 8$, resolving an open problem of Thomas and Akhtar. Our main structural result is a lift theorem: if the most balanced layer is a core, then so is the full hypergraph. Its proof shows that endomorphisms preserve the balanced layer and uses a separation argument to extend rigidity from that layer to every vertex. We then prove that the balanced layer is a core for every even n . For $n \equiv 2 \pmod{4}$, a symmetry reduction and a divisibility obstruction to a Steiner system exclude the only possible proper core. For $n \equiv 0 \pmod{4}$ and $n \geq 16$, link-clique bounds together with explicit combinatorial constructions rule out noninjective endomorphisms; $n = 12$ is handled by a separate certificate, while $n = 8$ is the known base case. Combining the even result with the odd balanced-layer theorem proved in a companion manuscript gives the result for all $n \geq 8$.

Keywords. Qualitative independence; hypergraph core; hypergraph homomorphism; covering array; merged Johnson graph; Steiner system.

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1 Introduction

A finite hypergraph is a *core* if every endomorphism is an automorphism; see [3] for the graph setting. In a binary qualitative-independence hypergraph, a vertex is an unordered bipartition of an n -element ground set, and three vertices form an edge when their common refinement has all eight cells nonempty. These hypergraphs encode triples of columns in binary covering arrays [5, 1].

Write $H_n = 3\text{-QI}(n, 2)$, and let B_n be the subhypergraph induced by the most balanced bipartitions. Thus B_n is $3\text{-UQI}(n, 2)$ when n is even and $3\text{-AUQI}(n, 2)$ when n is odd. Thomas and Akhtar proved that $H_8 = B_8$ is a core and asked for the core of H_n for $n > 8$ [8]. They also identified the 2-sections of the balanced layers with merged Johnson graphs.

For odd $n \geq 9$, the companion work [4] determines the endomorphism monoid of the corresponding merged Johnson graph and thereby shows that the balanced layer B_n is a core. The present paper treats the even balanced layers and establishes the passage from B_n to H_n .

Our lift theorem shows that endomorphisms preserve the balanced layer and that rigidity on this layer extends to the full hypergraph. Maximum link cliques identify the balanced layer intrinsically, while a separation lemma forces an endomorphism that fixes the layer pointwise to fix every vertex. For the even balanced layers, we analyse their 2-sections, using a Steiner-system obstruction in one

congruence class and a direct collision argument in the other. Together with the odd balanced-layer theorem, this resolves the problem of Thomas and Akhtar for every $n > 8$.

Theorem 1.1. *For every integer $n \geq 8$, the hypergraph H_n is a core.*

Theorem 1.2 (Even balanced layers). *For every even integer $n \geq 8$, the balanced layer $B_n = 3\text{-UQI}(n, 2)$ is a core.*

Theorem 1.3 (Balanced-layer lift). *Let $n \geq 8$. If the balanced layer B_n of H_n is a core, then H_n is a core.*

2 Qualitative independence and cores

Let Ω be an n -element set. A binary string is identified, up to interchanging its two symbols, with an unordered bipartition

$$[A] = \{A, \Omega \setminus A\}.$$

For a subset A of the current ground set, write A^c for its complement. Thus, for the present set Ω , $A^c = \Omega \setminus A$. Write $A^1 = A$ and $A^0 = A^c$. When an orientation has been fixed, we occasionally denote the vertex $[A]$ by its chosen representative A . Throughout, a homomorphism of simple 3-uniform hypergraphs is a vertex map taking every 3-edge to a 3-edge; an endomorphism has the same source and target. In the standard definition of $3\text{-QI}(n, 2)$, each binary symbol occurs at least $4 = 2^{3-1}$ times; equivalently, both blocks have size at least 4 [8]. Three vertices $[A], [B], [C]$ form an edge when all eight cells

$$A^{\epsilon_1} \cap B^{\epsilon_2} \cap C^{\epsilon_3}, \quad (\epsilon_1, \epsilon_2, \epsilon_3) \in \{0, 1\}^3,$$

are nonempty. This condition is independent of the chosen orientations.

For $n \geq 8$, the *balanced layer* (also called the *middle layer*) B_n is the subhypergraph induced by the bipartitions with block sizes

$$\left\lfloor \frac{n}{2} \right\rfloor, \quad \left\lceil \frac{n}{2} \right\rceil.$$

It is $3\text{-UQI}(n, 2)$ for even n and $3\text{-AUQI}(n, 2)$ for odd n .

The 2-section $[H]_2$ of a 3-uniform hypergraph H is the graph on $V(H)$ in which two vertices are adjacent when they lie together in a hyperedge.

Lemma 2.1 (Graph-to-hypergraph transfer). *Let H be a finite simple 3-uniform hypergraph. If $[H]_2$ is a core, then H is a core.*

Proof. Every endomorphism of H is an endomorphism of $[H]_2$. If $[H]_2$ is a core, that map is injective. On a finite vertex set, it is bijective. The induced map on 3-subsets is then a permutation, and it maps the finite edge set injectively into itself, hence onto itself. $\square$

3 Crossing cuts and the balanced-layer lift

Throughout this section, $n \geq 8$.

For a set T of size t , let $\text{Cross}(t)$ be the graph whose vertices are nontrivial unordered bipartitions of T , two of which are adjacent when all four cells of their common refinement are nonempty.

Proposition 3.1. *For $t \geq 4$,*

$$\omega(\text{Cross}(t)) = q(t) := \binom{t-1}{\lfloor t/2 \rfloor - 1}.$$

Moreover, $q(t+1) > q(t)$ for every $t \geq 4$.

Proof. For each cut, choose the side containing a fixed point p , and then delete p from that side. The resulting traces form an antichain in $T \setminus \{p\}$, because the chosen sides of two crossing cuts are incomparable. If $t = 2r$, Sperner's theorem therefore gives the upper bound $\binom{2r-1}{r-1}$, attained by the cuts whose chosen sides are the r -subsets containing p [7].

Now let $t = 2r + 1$, and write $U = T \setminus \{p\}$. Crossing also implies that $F \cup F' \neq U$ for any two distinct traces F and F' . Hence their complements in U form an intersecting antichain. Milner's theorem gives the upper bound $\binom{2r}{r-1}$, again attained by the cuts whose chosen sides are the r -subsets containing p [6]. Finally, the ratios

$$\frac{q(2r+1)}{q(2r)} = \frac{2r}{r+1} > 1, \quad \frac{q(2r+2)}{q(2r+1)} = \frac{2r+1}{r} > 1$$

give strict monotonicity. □

For a finite simple 3-uniform hypergraph H and $T \in V(H)$, let $L_H(T)$ denote the link graph:

$$R \sim_{L_H(T)} S \iff \{T, R, S\} \in E(H).$$

For a bipartition T , put

$$\tau(T) := \min\{|C| : C \in T\}.$$

Lemma 3.2 (Middle-layer invariance). *Every endomorphism of H_n maps B_n into B_n .*

Proof. We first prove that $\omega(L_{H_n}(T)) \leq q(\tau(T))$. The assertion is immediate for a clique of order at most one. For a larger clique $\mathcal{C}$, associate to each $R \in \mathcal{C}$ the unordered cut induced by R on a block of T of size $\tau(T)$. For $R, S \in \mathcal{C}$, the four cells of these local cuts are precisely the four cells of the triple (T, R, S) inside that block. Thus the local cuts cross. The restriction is injective, since equal unordered cuts would leave two of those cells empty. Therefore

$$\omega(L_{H_n}(T)) \leq q(\tau(T)).$$

If $T \in B_n$, let $b = \lfloor n/2 \rfloor$. On each block, choose the extremal family from Proposition 3.1, orienting every chosen side towards its fixed point. Retain all $q(b)$ cuts on the b -block and $q(b)$ cuts on the other block; when n is odd, discard the surplus cuts from the $q(b+1)$ -family. Pair the two local families arbitrarily and take the unions of corresponding oriented sides. Every pair of resulting global cuts realises all four joint colours inside each block of T , and hence forms an edge with T . Each local side has size at least 2, so both blocks of every global cut have size at least 4; the fixed orientations also make the global cuts distinct. We have constructed a $q(b)$ -clique in $L_{H_n}(T)$, including the boundary cases $n = 8, 9$.

An endomorphism maps every link clique of size at least two injectively into the link at the image vertex. Thus, for $T \in B_n$,

$$q(b) = \omega(L_{H_n}(T)) \leq \omega(L_{H_n}(f(T))) \leq q(\tau(f(T))).$$

Strict monotonicity and $\tau(f(T)) \leq b$ imply $\tau(f(T)) = b$. □

Lemma 3.3 (Separation by the balanced layer). *For distinct $X, Y \in V(H_n)$, there exist $R, S \in B_n$ such that*

$$\{X, R, S\} \in E(H_n), \quad \{Y, R, S\} \notin E(H_n).$$

Proof. Write the two blocks of X as X_0 and X_1 . Since $X \neq Y$, some block Y_* of Y meets both blocks of X ; otherwise the two partitions coincide. Put $D = Y_*^c$.

Colour Ω with the four colours 00, 01, 10, 11. Choose one point in each $X_i \cap Y_*$ and colour those two points 11. In each X_i , colour three additional points 00, 01, 10. Every X_i now contains all four colours, while D contains no point of colour 11.

For $n = 2k$, complete the colour multiplicities to

$$(c_{00}, c_{01}, c_{10}, c_{11}) = (2, k - 2, k - 2, 2);$$

for $n = 2k + 1$, complete them to

$$(3, k - 2, k - 2, 2).$$

The eight points already chosen contribute two occurrences of each colour. Since $k \geq 4$, the displayed targets dominate these initial counts and sum to n , so the remaining points can be coloured to achieve them. Define

$$R^* = \{x : c(x) \in \{10, 11\}\}, \quad S^* = \{x : c(x) \in \{01, 11\}\}, \quad R = [R^*], \quad S = [S^*].$$

The displayed multiplicities give $|R^*| = |S^*| = \lfloor n/2 \rfloor$, so R and S belong to B_n . Every block of X sees all four joint colours, whereas $D \cap R^* \cap S^* = \emptyset$. This proves the two claimed incidences. $\square$

Proof of Theorem 1.3. Let $f \in \text{End}(H_n)$. By Lemma 3.2, $g = f|_{B_n}$ is an endomorphism of B_n , hence an automorphism. Let d be its finite order and set $h = f^d$. Then h fixes B_n pointwise.

If $h(X) = Y \neq X$, choose $R, S \in B_n$ from Lemma 3.3. Applying h would map the edge $\{X, R, S\}$ to the nonedge $\{Y, R, S\}$, a contradiction. Hence $f^d = h = \text{id}$, so f is bijective and its inverse is the endomorphism f^{d-1} . $\square$

4 Even balanced layers with $n \equiv 2 \pmod{4}$

Let $n = 2k$, where $k \geq 5$ is odd, and let $\widehat{\Omega}$ be a $2k$ -element ground set. Put

$$B_{2k} = 3\text{-UQI}(2k, 2), \quad G_{2k} = [B_{2k}]_2.$$

The vertices are unordered equipartitions $[A] = \{A, A^c\}$, with $|A| = k$. If $s = |A \cap B|$, the four intersection cells, in the order

$$A \cap B, \quad A \setminus B, \quad B \setminus A, \quad \widehat{\Omega} \setminus (A \cup B),$$

have sizes $s, k - s, k - s, s$. A third equipartition, represented by a k -set C , splits all four cells if and only if

$$2 \leq s \leq k - 2.$$

Necessity follows because each cell must contain a point of C and a point outside C , and hence must have size at least 2. For sufficiency, choose C so that its intersections with the four cells have respective sizes

$$1, \quad 1, \quad k - s - 1, \quad s - 1.$$

Write the ground set as $\widehat{\Omega} = \Omega \cup \{\infty\}$, with $|\Omega| = 2k - 1$. Orient each equipartition towards the block containing ∞ , and delete ∞ . This identifies the vertices bijectively with the r -subsets of Ω , where

$$r = k - 1, \quad |\Omega| = 2r + 1 = 2k - 1.$$

If two reduced representatives meet in j points, their corresponding oriented k -blocks meet in $j + 1$ points. Thus distinct vertices are nonadjacent in G_{2k} precisely when $j \in \{0, r - 1\}$. Let Γ be the graph on $\binom{\Omega}{r}$ in which two sets are adjacent exactly in those two cases. Then

$$G_{2k} = \overline{\Gamma}.$$

Lemma 4.1 (Clique number of Γ). *Γ is vertex-transitive and $\omega(\Gamma) = r + 2$.*

Proof. The natural action of S_Ω on r -subsets is transitive and preserves Γ . For the lower bound, fixing an $(r - 1)$ -subset gives a clique of size $r + 2$. For the upper bound, first suppose that a clique contains disjoint A, B . There is one point x outside $A \cup B$. If C is adjacent to both, then

$$|C \cap A| + |C \cap B| + \mathbf{1}_{x \in C} = r,$$

with both intersections in $\{0, r - 1\}$. Hence C is

$$(A \setminus \{a\}) \cup \{x\} \quad \text{or} \quad (B \setminus \{b\}) \cup \{x\}.$$

One set of each type meets in the singleton $\{x\}$, so the two types cannot be mixed when $r \geq 3$. The clique has at most $2 + r = r + 2$ members.

Now suppose that there is no disjoint pair. All pairwise intersections have size $r - 1$. Write

$$A = D \cup \{a\}, \quad B = D \cup \{b\}, \quad |D| = r - 1.$$

Indeed, a set C meeting both A and B in $r - 1$ points either contains D , or omits some $d \in D$. In the latter case the two intersection conditions force $C = (D \setminus \{d\}) \cup \{a, b\} \subseteq D \cup \{a, b\}$. Every further member either contains D , or is an r -subset of $D \cup \{a, b\}$. If only the first type occurs, the family is contained in the $r + 2$ sets obtained by adjoining one point to D . If a member of the second type occurs, it is not adjacent in Γ to any set $D \cup \{z\}$ with $z \notin D \cup \{a, b\}$, and the clique is contained in the $r + 1$ r -subsets of $D \cup \{a, b\}$. This proves the bound. $\square$

Recall that a Steiner system $S(t, k, v)$ is a family of k -subsets of a v -set in which every t -subset lies in exactly one block.

Lemma 4.2 (Design implication). *If*

$$\chi(\overline{\Gamma}) = \omega(\overline{\Gamma}),$$

then a Steiner system $S(r - 1, r, 2r + 1)$ exists.

Proof. Let $\mathcal{Q}$ be a clique and $\mathcal{I}$ an independent set of the vertex-transitive graph Γ . Averaging $|g\mathcal{Q} \cap \mathcal{I}| \leq 1$ over a transitive automorphism group gives

$$|\mathcal{Q}||\mathcal{I}| \leq \binom{2r + 1}{r}.$$

Assume $\chi(\overline{\Gamma}) = \omega(\overline{\Gamma}) = c$, and put $N = \binom{2r + 1}{r}$. Since a colour class of $\overline{\Gamma}$ is a clique of Γ , Lemma 4.1 gives $N \leq c(r + 2)$. On the other hand, $c = \omega(\overline{\Gamma}) = \alpha(\Gamma)$, so the clique-coclique inequality above gives $c(r + 2) \leq N$. Therefore equality holds throughout:

$$c = \frac{\binom{2r + 1}{r}}{r + 2}$$

and every colour class has size $r + 2$.

Take a maximum clique $\mathcal{B}$ of $\bar{\Gamma}$. It is an independent set in Γ , so no two of its r -subsets share an $(r - 1)$ -subset. Moreover,

$$|\mathcal{B}|r = \frac{\binom{2r+1}{r}}{r+2}r = \binom{2r+1}{r-1}.$$

Thus every $(r - 1)$ -subset lies in exactly one member of $\mathcal{B}$, so $\mathcal{B}$ is the block set of a Steiner system $S(r - 1, r, 2r + 1)$. $\square$

Proposition 4.3. *If $k \geq 5$ is odd, then 3-UQI($2k, 2$) is a core.*

Proof. The group $S_{\hat{\Omega}} \cong S_{2k}$ acts on unordered equipartitions and preserves G_{2k} . By the adjacency criterion above, a distinct nonedge has $s \in \{1, k - 1\}$; the cases $s = 0$ and $s = k$ represent the same unordered equipartition. After choosing orientations, the four intersection cells therefore have sizes $1, k - 1, k - 1, 1$. The alternatives $s = 1$ and $s = k - 1$ describe the same unordered pair after complementing one chosen block. Hence $S_{\hat{\Omega}}$, and therefore $\text{Aut}(G_{2k})$, is transitive on unordered nonedges. Corollary 2.2 of Cameron and Kazanidis implies that G_{2k} is either a core or a graph with a complete core [2]. A finite graph has a complete core exactly when its clique and chromatic numbers agree.

By Lemma 4.2, equality of the clique and chromatic numbers would yield

$$S(k - 2, k - 1, 2k - 1).$$

In such a design, the number of blocks through a fixed $(k - 3)$ -subset is

$$\lambda_{k-3} = \frac{\binom{k+2}{1}}{\binom{2}{1}} = \frac{k+2}{2},$$

which is not an integer for odd k . Hence the complete-core alternative is impossible, so G_{2k} is a core. Lemma 2.1 now shows that B_{2k} is a core. $\square$

5 Even balanced layers with $n \equiv 0 \pmod{4}$

Let $n = 4m$ and put $B_{4m} = 3\text{-UQI}(4m, 2)$. Write the $4m$ -element ground set as $\hat{\Omega} = \Omega \cup \{\infty\}$, where $|\Omega| = 4m - 1$. Orient each equipartition towards its unique block containing ∞ . Removing ∞ from that block gives a bijection with the fixed-point model

$$V(B_{4m}) = \binom{\Omega}{2m-1}, \quad |\Omega| = 4m - 1.$$

Here, the bit 1 denotes membership in the corresponding reduced set, equivalently membership in its oriented block containing ∞ . For three vertices, the 111-cell contains ∞ automatically, so they form an edge precisely when the other seven Boolean cells in Ω are nonempty.

Put $r = 2m - 1$ and $a = |X \cap Y|$. In the order

$$X \cap Y, \quad X \setminus Y, \quad Y \setminus X, \quad \Omega \setminus (X \cup Y),$$

the four pair regions have sizes $a, r - a, r - a, a + 1$. If $1 \leq a \leq r - 2$, choose a third r -set whose intersections with these four regions have respective sizes

$$0, \quad 1, \quad r - a - 1, \quad a.$$

The first region need only contribute to its non-111 cell, whereas each of the other three regions is split between the two values of the third bit. This makes all seven required cells nonempty. Conversely, the non-111 cell in the first region and the two cells in each difference region force $a \geq 1$ and $r - a \geq 2$. Thus the 2-section joins X and Y exactly when

$$1 \leq |X \cap Y| \leq 2m - 3.$$

Distinct nonedges therefore have intersection 0 or $r - 1$. In the equipartition model these belong to the same orbit: complementing one oriented block interchanges the two cases, and the four cells have sizes $1, r, r, 1$. Hence all unordered nonedges form one orbit under the action of $S_{\widehat{\Omega}} \cong S_{4m}$ transported to the fixed-point model.

Lemma 5.1 (A collision criterion). *Suppose that H is a vertex-transitive 3-uniform hypergraph, every link of H has clique number at most M , and $\text{Aut}(H)$ is transitive on the unordered pairs of distinct vertices that are nonedges of $[H]_2$. Suppose also that a fixed nonedge $\{P, Q\}$ admits distinct vertices $R_1, \dots, R_{M+1}$ such that, for every $i < j$, at least one of*

$$\{P, R_i, R_j\}, \quad \{Q, R_i, R_j\}$$

is an edge. Then H is a core.

Proof. If an endomorphism identifies P and Q , the images of $R_1, \dots, R_{M+1}$ are distinct: for each $i < j$, the pair $\{R_i, R_j\}$ is contained in a hyperedge and hence its vertices cannot have the same image. They form an $(M+1)$ -clique in the link of the common image, a contradiction. More generally, two distinct vertices identified by an endomorphism must be nonadjacent in the 2-section. By the orbit hypothesis an automorphism sends the unordered pair $\{P, Q\}$ to that pair, and precomposition reduces the collision to the fixed pair $\{P, Q\}$. Thus no collision is possible. $\square$

Proposition 5.2. *For every $m \geq 4$, 3-UI(4m, 2) is a core.*

Proof. The action of $S_{\widehat{\Omega}}$ is transitive on $V(B_{4m})$. For a clique $\mathcal{C}$ in $L_{B_{4m}}(T)$, the traces $\{T \cap R : R \in \mathcal{C}\}$ form an antichain in 2^T : a containment between two traces would make one of the two difference cells inside T empty. Hence every link of B_{4m} has clique number at most

$$M_m := \binom{2m-1}{m-1}.$$

Construction. We construct M_m central vertices indexed by the $(m-1)$ -subsets of Q , and then add one exceptional vertex. The three types below are chosen so that pairs of central vertices are witnessed by Q , whereas pairs containing the exceptional vertex are witnessed by either P or Q .

Choose pairwise disjoint sets K , $\{p, q\}$, and W with

$$|K| = 2m - 2, \quad |W| = 2m - 1,$$

and put $P = K \cup \{p\}$, $Q = K \cup \{q\}$. Fix $x \in K$ and distinct $u, v \in W$, and put $O = \{p\} \cup W$. The pair $\{P, Q\}$ is a nonedge because $|P \cap Q| = 2m - 2$.

Divide the family $\binom{Q}{m-1}$ into three types, labelled 0, 10, 11, respectively:

$$\begin{array}{l|l} 0 & x \notin A, \\ 10 & x \in A, \ q \notin A, \\ 11 & x, q \in A. \end{array}$$

The complement pairs of m -subsets of O likewise fall into three classes: either u and v are separated, or they lie together in the unique p -free representative, or they lie together in its complement. Thus these classes exhaust all complement pairs. Their cardinalities are

type	condition	number
0	u, v separated	$\binom{2m-2}{m-1}$
10	p -free representative contains u, v	$\binom{2m-3}{m-2}$
11	p -free representative avoids u, v	$\binom{2m-3}{m-3}$.

The three subfamilies of $\binom{Q}{m-1}$ have, respectively, the three cardinalities displayed above: one chooses $m-1$ points while avoiding x , chooses x and $m-2$ further points while avoiding q , or chooses x, q and $m-3$ further points. For each type, fix a bijection between the two corresponding classes.

For types 10 and 11, let C_A be the unique p -free representative of the assigned complement pair; for type 0, choose either representative. Set $R_A = A \cup C_A$. Since $Q \cap O = \emptyset$, each R_A has size $(m-1) + m = 2m-1$ and hence is a vertex of B_{4m} .

Pairs of central vertices. For distinct A and A' , put

$$s = |A \cap A'|, \quad t = |C_A \cap C_{A'}|.$$

Since A and A' are distinct and have the same size, $0 \leq s \leq m-2$. Since their assigned complement pairs are distinct, C_A and $C_{A'}$ are neither equal nor complementary, and $1 \leq t \leq m-1$. With the non-111 Boolean cells indexed in the order 000, 001, 010, 011, 100, 101, 110, their sizes for $(Q, R_A, R_{A'})$ are

	000	001	010	011	100	101	110
size	t	$m-t$	$m-t$	t	$s+1$	$m-1-s$	$m-1-s$.

They are all positive. Hence, for distinct A, A' ,

$$\{Q, R_A, R_{A'}\} \in E(B_{4m}).$$

Pairs containing the exceptional vertex. Put

$$Z = \{x, p\} \cup (W \setminus \{u, v\}).$$

For A of a given type, let $\text{anc}(A) \in \{P, Q\}$ be the anchor shown in the corresponding row below. The seven non-111 Boolean-cell sizes for $(\text{anc}(A), Z, R_A)$ are

type	anchor	000	001	010	011	100	101	110
0	Q	1	1	$m-1$	$m-1$	$m-1$	$m-1$	1
10	P	1	2	$m-1$	$m-2$	$m-1$	$m-2$	1
11	P	2	1	$m-3$	m	m	$m-3$	1.

All are positive for $m \geq 4$, and hence $\{\text{anc}(A), Z, R_A\} \in E(B_{4m})$.

The family

$$\{Z\} \cup \left\{ R_A : A \in \binom{Q}{m-1} \right\}$$

has $M_m + 1$ vertices. Every pair of central vertices is witnessed by Q , and every pair consisting of Z and a central vertex is witnessed by the anchor in the table. The hypotheses of Lemma 5.1 are therefore satisfied, and B_{4m} is a core. $\square$

5.1 The exceptional value $n = 12$

When $m = 3$, the two entries $m - 3$ in the type 11 row vanish. We instead use an obstruction with eleven auxiliary vertices. After deleting ∞ , let

$$\Omega_{12}^\circ := \{0, 1, \dots, 10\},$$

and use the model $V(B_{12}) = \binom{\Omega_{12}^\circ}{5}$. Set

$$P = \{0, 1, 2, 3, 4\}, \quad Q = \{0, 1, 2, 3, 5\}.$$

The eleven auxiliary vertices and their anchor table are given in Appendix A; the accompanying program `supplementary/verify_certificates.py` verifies all 55 required pairwise incidences. Every link $L_{B_{12}}(T)$ has clique number at most $\binom{5}{2} = 10$, while identifying P and Q would force an 11-clique in one link. The preceding vertex-transitivity argument and the single-orbit property for nonedges also apply when $m = 3$. Hence Lemma 5.1 applies to this certificate and proves that $B_{12} = 3\text{-UQI}(12, 2)$ is a core.

Proof of Theorem 1.2. For $n = 8$, all admissible vertices have block sizes $4 + 4$, so $3\text{-QI}(8, 2) = 3\text{-UQI}(8, 2)$; this case is due to Thomas and Akhtar [8, Theorem 4]. Proposition 4.3 treats $n \equiv 2 \pmod{4}$, the preceding subsection treats $n = 12$, and Proposition 5.2 treats $n \equiv 0 \pmod{4}$, $n \geq 16$. $\square$

Proof of Theorem 1.1. For even n , the balanced layer is a core by Theorem 1.2. For odd $n \geq 9$, the balanced layer is a core by Theorem 7.1 and Corollary 7.2 of the companion manuscript [4]. Theorem 1.3 therefore proves the result. The case $n = 8$ is already included in the even statement. $\square$

A The $n = 12$ collision certificate

The auxiliary vertices $R_1, \dots, R_{11}$ are

R_1	$\{1, 2, 3, 5, 10\}$	R_2	$\{0, 3, 8, 9, 10\}$
R_3	$\{0, 2, 7, 9, 10\}$	R_4	$\{2, 3, 4, 9, 10\}$
R_5	$\{1, 3, 4, 8, 10\}$	R_6	$\{1, 2, 4, 7, 10\}$
R_7	$\{0, 5, 7, 8, 9\}$	R_8	$\{0, 1, 6, 7, 10\}$
R_9	$\{3, 4, 5, 7, 9\}$	R_{10}	$\{2, 4, 5, 6, 9\}$
R_{11}	$\{1, 4, 5, 7, 8\}$		

and the upper-triangular anchor table is

	2	3	4	5	6	7	8	9	10	11
1	P	P	P	P	P	P	P	P	P	P
2		P	Q	Q	Q	Q	P	P	P	P
3			Q	Q	Q	Q	P	P	P	P
4				P	P	P	Q	Q	Q	Q
5					P	P	P	Q	Q	Q
6						P	Q	P	Q	Q
7							Q	Q	P	Q
8								P	P	P
9									P	P
10										P

An entry P or Q indicates that the corresponding pair of auxiliary vertices forms a hyperedge with that anchor. Each incidence is a direct seven-cell check; for example, the entry in row 1, column 2 states that, in the order 000, 001, 010, 011, 100, 101, 110, the seven required cell sizes for (P, R_1, R_2) are $(2, 2, 1, 1, 1, 1, 2)$. The accompanying script verifies all 55 incidences directly from the listed sets.

Declaration on generative AI

ChatGPT assisted with exploratory calculations, code development, drafting, and editorial review. The author is responsible for the mathematical statements, proofs, and accompanying code.

Data and code availability

The source, verification scripts, and input data are archived on Zenodo, version v0.1.0, at <https://doi.org/10.5281/zenodo.22164331>. The scripts check the finite $n = 12$ certificate, the boundary instance $n = 16$ of the general construction, and the stated small lift instances.

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